BBC LEARNING ENGLISH

6 Minute English

Could vaccines cure cancer?

This is not a word-for-word transcript.

Georgie
Hi Phil.

Phil
Hi Georgie.

Georgie
Phil, have you had many vaccines in your life? So, these are injections to protect you from diseases.

Phil
Oh, I've had loads. I had all the ones you have when you're a child, but I've also had some for travelling and working abroad so… yeah, I've had lots.

Georgie
OK. Well, many of us will have had vaccines to stop us getting common diseases such as flu or Covid 19, but could injections help us fight more complicated diseases too? In this episode of 6 Minute English, we'll be looking at some new research that has scientists very excited, and we'll be asking the question, 'Could vaccines cure cancer?' OK. Let's start with a quiz question, Phil. There are lots of different types of cancer affecting many parts of the body, but according to the World Health Organisation, what is the most common cancer worldwide? Is it:

a) lung cancer,
b) skin cancer, or
c) stomach cancer?

Phil
Well, I don't know. I'm going to guess b) skin cancer.

Georgie
Well, we'll find out the answer at the end of the programme.

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Phil
Right. Now, this idea of a vaccine that could cure cancer sounds really interesting. Could you tell us a bit more about it, Georgie?

Georgie
Well, scientists have been trying to develop a cancer vaccine for quite a long time. If you think about many existing cancer treatments, such as radiotherapy and chemotherapy, these have quite a few side effects. But the idea is that a cancer vaccine would help the body's natural immune system help fight the disease, which could have fewer side effects and also be more effective. Scientists are particularly hopeful about a new vaccine developed to treat skin cancer. BBC Radio 4 programme Inside Science has been hearing about the latest research. Cancer cell biologist Anne Willis spoke to presenter Tom Whipple.

Tom Whipple
I guess this moment has been quite **a long time coming**, hasn't it? There's been some **false dawns** in cancer vaccines.

Anne Willis
I mean, there's been a lot of work in this **space**, and it's all been very encouraging, particularly in the animal models, but getting this to translate into humans can be a much harder, longer process, so it's very exciting that there's this **breakthrough** – apparent **breakthrough** – that's come through today.

Phil
Tom says that the development of the cancer vaccine has been **a long time coming**, and this means that it's taken a lot of time to achieve.

Georgie
Yes. Tom also says there have been some **false dawns**, an expression meaning that something looks like it's improving when it's actually not.

Phil
Anne says there's been a lot of work in this **space**. **Space** here means field of research – so, in this context, research into cancer vaccines.

Georgie
Scientists think they might have made a **breakthrough**. That means they've made significant progress.

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Phil
And Georgie, these vaccines that we're talking about, they aren't quite like the ones that we're all familiar with, are they? This isn't a vaccine to prevent you from getting skin cancer, is it?

Georgie
No. The vaccine being tested is used to treat people after they've had a tumour removed. It's used to target any cancer cells that have been spread around the body, to stop the cancer from coming back, and the vaccine is made specially for each patient by looking at the cancerous cells in the tumour that's been removed. Anne Willis explained more to BBC Radio 4's Inside Science.

Anne Willis
So, cancer cells are very different from normal cells. They contain lots of mutations. So, you sequence the DNA of the cancer cell to work out how it's different from a normal cell. And then, particularly for the bits of the cancer that's different on the surface of the cell, you take those bits of DNA and then you put them into the RNA vaccine, and then they'll make these bits of proteins that get recognised by the immune system. So, it's a way of **priming** the immune system to the cancer.

Phil
Anne is saying that the scientists work out what in the cancer cells is different and put these bits into the vaccine.

Georgie
Right. The vaccine then becomes a way of **priming** the immune system to cancer. If you **prime** something, you get it ready. The vaccine basically helps the immune system get better at recognising and fighting cancer cells in the body.

Phil
Now, that sounds like some really clever science here. Could it make a big difference, Georgie?

Georgie
Well, more trials are needed, but scientists are hopeful. Here's Anne Willis again talking to Tom Whipple, presenter of Inside Science.

Tom Whipple
Are you excited?

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Anne Willis
Oh, I mean, very! I'm just, you know… anything that will improve lives of cancer patients is always wonderful. And actually, seeing that this extraordinary technology is, sort of, starting to reach its **promise** – this is very exciting for everybody.

Phil
So, Anne is excited. She says that after a long time working on this technology, it's starting to reach its **promise**. And **promise** here means positive potential.

Georgie
OK. Phil, I think it's time I revealed the answer to my question. I asked you, "What is the most common form of cancer worldwide?" You said skin cancer, but the answer is actually lung cancer. In 2024, there were an estimated 2.4 million cases worldwide, according to the World Health Organisation.

Phil
OK. Let's recap the vocabulary we learned in this programme, starting with the expression **a long time coming**, which means something has taken a long time to achieve.

Georgie
A **false dawn** is when something appears to be improving, when it actually isn't.

Phil
Space can mean field of research.

Georgie
A **breakthrough** refers to making significant progress with something.

Phil
If you **prime** something, you get it ready.

Georgie
And **promise** can mean positive potential.

Phil
Once again, our six minutes are up. Test what you've learned with the worksheet on our website. And that's bbclearningenglish.com.

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Georgie
Thanks for joining us and goodbye.

Phil
Bye.

VOCABULARY

a long time coming
taking a long time to achieve

false dawn
situation where something appears to be improving, when actually it isn't

space
field of research

breakthrough
significant progress in something

prime (something)
get (something) ready

promise
positive potential

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